

BAZIL UDDIN KHAN

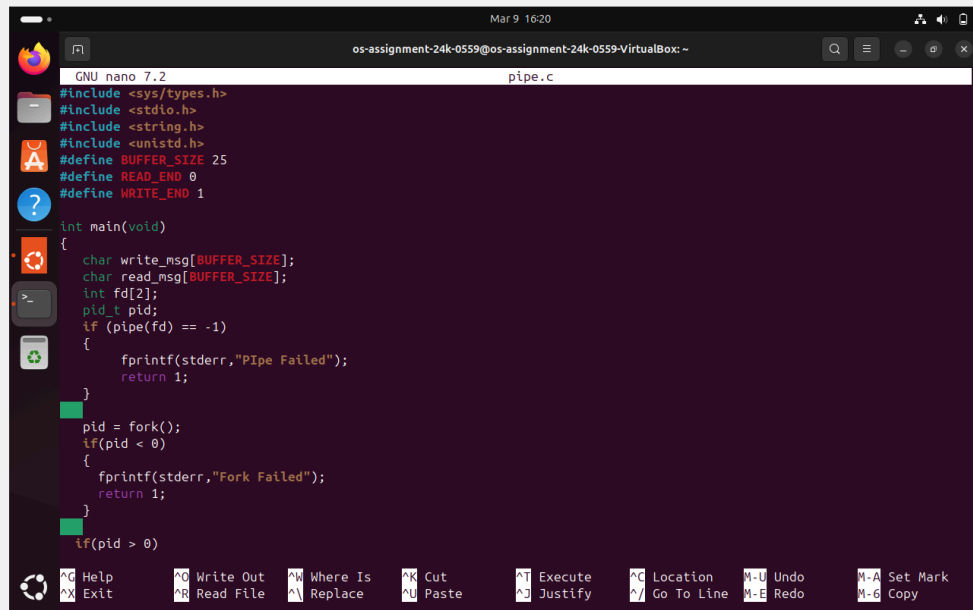
LAB 06

EXPEIREMENT 6A:

IN LAB:

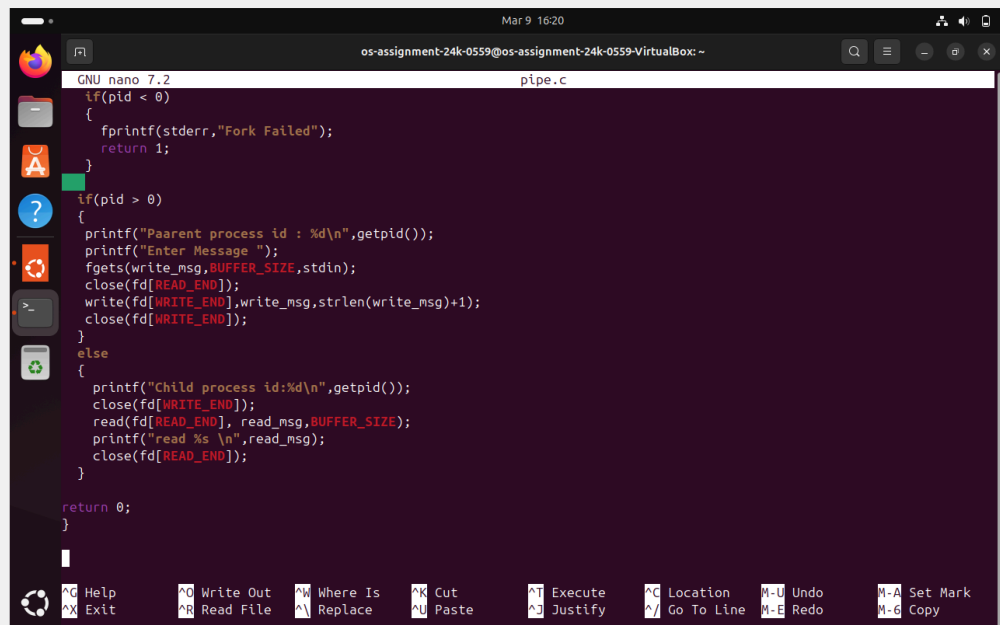
Ubuntu_Assignment [Running] - Oracle VirtualBox
File Machine View Input Devices Help

```
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: ~
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano pipe.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gedit pipe.c
Command 'gedit' not found, but can be installed with:
sudo snap install gedit # version 48.1, or
sudo apt install gedit # version 46.1-3
See 'snap info gedit' for additional versions.
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano pipe.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc pipe.c -o p1
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./p1
read Greetingsos-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano p
ipe.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc pipe.c -o p1
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./p1
read Greetings
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano pipe.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc pipe.c -o p1
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./p1
Paarent process id : 3693
Enter Message Child process id:3694
HelloToOs
read HelloToOs
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$
```



```
GNU nano 7.2 pipe.c
#include <sys/types.h>
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#define BUFFER_SIZE 25
#define READ_END 0
#define WRITE_END 1

int main(void)
{
    char write_msg[BUFFER_SIZE];
    char read_msg[BUFFER_SIZE];
    int fd[2];
    pid_t pid;
    if (pipe(fd) == -1)
    {
        fprintf(stderr, "Pipe Failed");
        return 1;
    }
    pid = fork();
    if (pid < 0)
    {
        fprintf(stderr, "Fork Failed");
        return 1;
    }
    if (pid > 0)
```



```

    if (pid < 0)
    {
        fprintf(stderr, "Fork Failed");
        return 1;
    }
    if (pid > 0)
    {
        printf("Parent process id : %d\n", getpid());
        printf("Enter Message ");
        fgets(write_msg, BUFFER_SIZE, stdin);
        close(fd[READ_END]);
        write(fd[WRITE_END], write_msg, strlen(write_msg)+1);
        close(fd[WRITE_END]);
    }
    else
    {
        printf("Child process id: %d\n", getpid());
        close(fd[WRITE_END]);
        read(fd[READ_END], read_msg, BUFFER_SIZE);
        printf("read %s \n", read_msg);
        close(fd[READ_END]);
    }

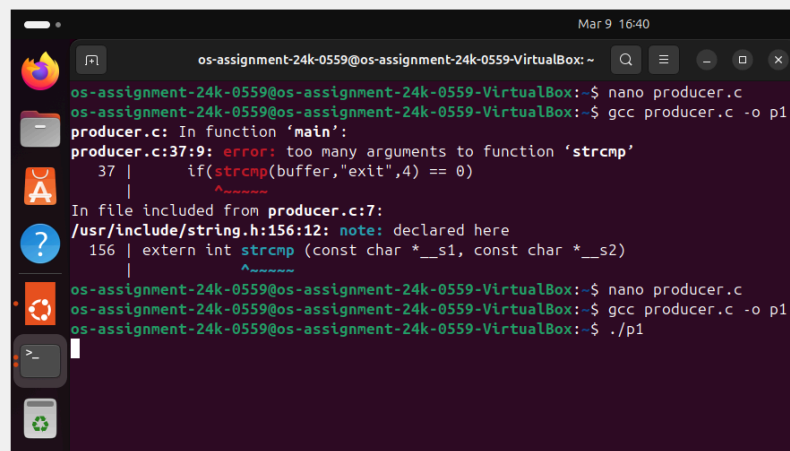
    return 0;
}
```

EXPEIREMENT 06 B).

IN LAB TASKS).

Ubuntu_Assignment [Running] - Oracle VirtualBox

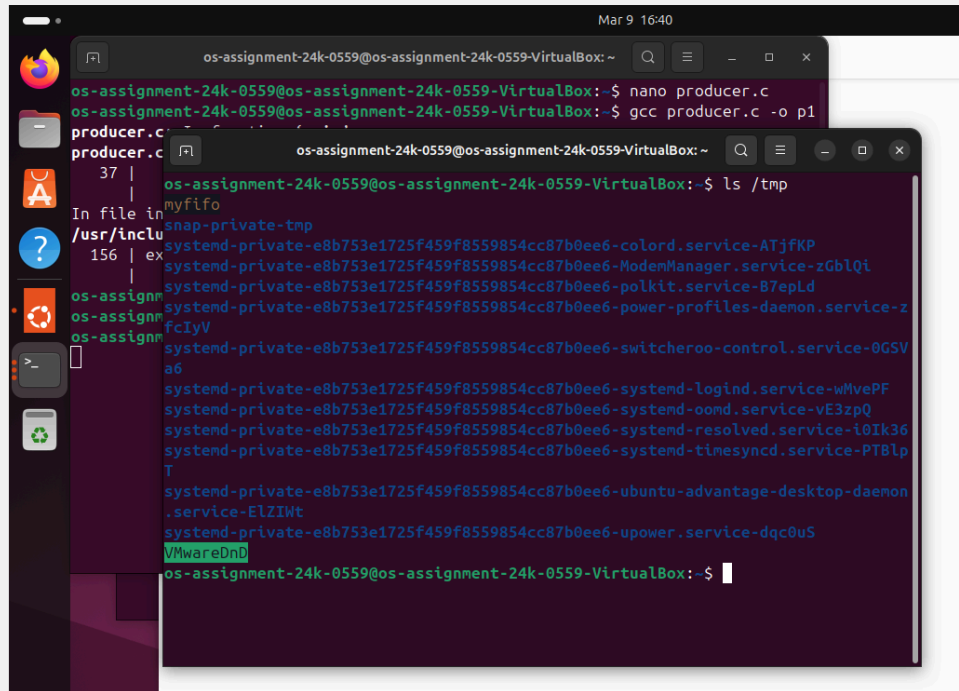
File Machine View Input Devices Help



```
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: ~
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano producer.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc producer.c -o p1
producer.c: In function 'main':
producer.c:37:9: error: too many arguments to function 'strcmp'
   37 |         if(strcmp(buffer,"exit",4) == 0)
      |         ^~~~~~
In file included from producer.c:7:
/usr/include/string.h:156:12: note: declared here
   156 | extern int strcmp (const char *__s1, const char *__s2)
      |
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano producer.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc producer.c -o p1
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./p1
```

Ubuntu_Assignment [Running] - Oracle VirtualBox

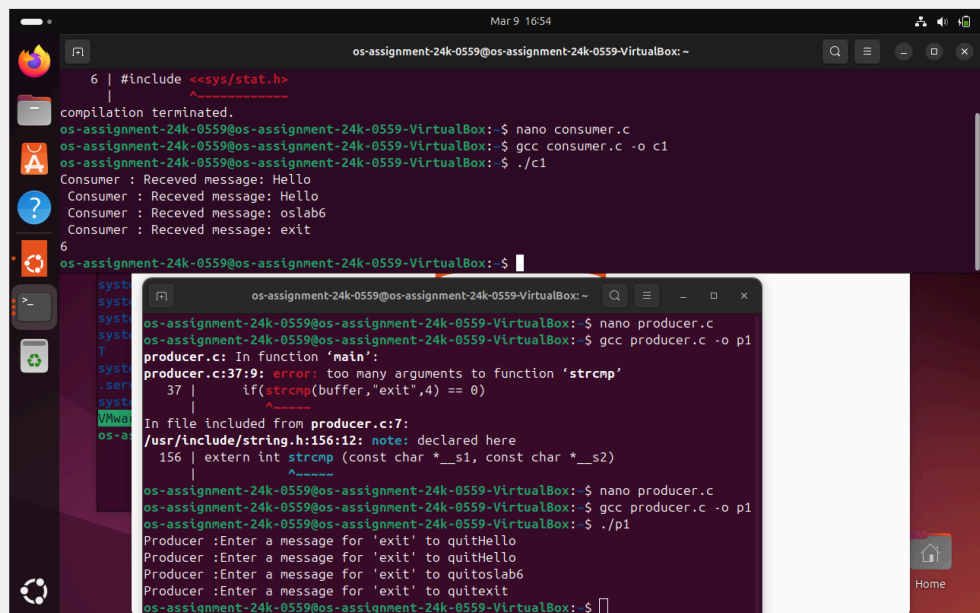
File Machine View Input Devices Help



```
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: ~  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $ nano producer.c  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $ gcc producer.c -o p1  
producer.c  
producer.c  
37 |  
In file included from producer.c:156:  
/usr/include/string.h:156:12: note: declared here  
extern int strcmp(const char *__s1, const char *__s2)  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $ ls /tmp  
myfifo  
snap-private-tnp  
systemd-private-e8b753e1725f459f8559854cc87b0ee6-color.service-ATjfkP  
systemd-private-e8b753e1725f459f8559854cc87b0ee6-ModemManager.service-zGblQi  
systemd-private-e8b753e1725f459f8559854cc87b0ee6-polkit.service-B7epLd  
systemd-private-e8b753e1725f459f8559854cc87b0ee6-power-profiles-daemon.service-z  
fclyV  
systemd-private-e8b753e1725f459f8559854cc87b0ee6-switcheroo-control.service-0GSV  
a6  
systemd-private-e8b753e1725f459f8559854cc87b0ee6-systemd-logind.service-wMvePF  
systemd-private-e8b753e1725f459f8559854cc87b0ee6-systemd-oond.service-vE3zpQ  
systemd-private-e8b753e1725f459f8559854cc87b0ee6-systemd-resolved.service-i0Ik36  
systemd-private-e8b753e1725f459f8559854cc87b0ee6-systemd-timesyncd.service-PTBlp  
T  
systemd-private-e8b753e1725f459f8559854cc87b0ee6-ubuntu-advantage-desktop-daemon  
.service-ELZIwt  
systemd-private-e8b753e1725f459f8559854cc87b0ee6-upower.service-dqc0uS  
VMwareDnD  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $
```

Ubuntu_Assignment [Running] - Oracle VirtualBox

File Machine View Input Devices Help



```
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: ~  
6 | #include <<sys/stat.h>  
|  
compilation terminated.  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $ nano consumer.c  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $ gcc consumer.c -o c1  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $ ./c1  
Consumer : Received message: Hello  
Consumer : Received message: Hello  
Consumer : Received message: oslab6  
Consumer : Received message: exit  
6  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $ nano producer.c  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $ gcc producer.c -o p1  
producer.c: In function 'main':  
producer.c:37:9: error: too many arguments to function 'strcmp'  
37 | if(strcmp(buffer,"exit",4) == 0)  
| ~~~~~  
In file included from producer.c:7:  
/usr/include/string.h:156:12: note: declared here  
156 | extern int strcmp(const char *__s1, const char *__s2)  
| ~~~~~  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $ nano producer.c  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $ gcc producer.c -o p1  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $ ./p1  
Producer :Enter a message for 'exit' to quitHello  
Producer :Enter a message for 'exit' to quitHello  
Producer :Enter a message for 'exit' to quitoslab6  
Producer :Enter a message for 'exit' to quitexit  
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: $
```

Mar 9 16:54

os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: ~

GNU nano 7.2 consumer.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <string.h>

#define FIFO_FILE "/tmp/myfifo"

int main()
{
    int fd;
    char buffer[BUFSIZ];
    ssize_t num_bytes;

    fd = open(FIFO_FILE, O_RDONLY);
    if(fd == -1)
    {
        perror("OPEN");
        exit(EXIT_FAILURE);
    }

    while(1)
    {
        num_bytes = read(fd,buffer,BUFSIZ);
        if(num_bytes > 0)
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo

```

GNU nano 7.2                                     consumer.c
ssize_t num_bytes;

fd = open(FIFO_FILE, O_RDONLY);
if(fd == -1)
{
    perror("OPEN");
    exit(EXIT_FAILURE);
}

while(1)
{
    num_bytes = read(fd,buffer,BUFSIZ);
    if(num_bytes > 0)
    {
        printf("Consumer : Received message: %s ",buffer);
    }

    if(strncmp(buffer, "exit",4) == 0)
    {
        break;
    }
}

close(fd);
return 0;
}
  
```

Mar 9 16:55

os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: ~

GNU nano 7.2 producer.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <string.h>

#define FIFO_FILE "/tmp/myFifo"

int main()
{
    int fd;
    char buffer[BUFSIZ];
    ssize_t num_bytes;

    mkfifo(FIFO_FILE,0666);
    fd = open(FIFO_FILE,O_WRONLY);
    if(fd == 1)
    {
        perror("OPen");
        exit(EXIT_FAILURE);
    }

    while(1)
    {
        printf("Producer :Enter a message for 'exit' to quit");
        fgets(buffer,BUFSIZ,stdin);
    }
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Un
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Re

os-assignment-24k-0559@os-assignment-24k-0559-VirtualB

GNU nano 7.2

```
producer.c
```

```
{
    perror("OPen");
    exit(EXIT_FAILURE);
}

while(1)
{
    printf("Producer :Enter a message for 'exit' to quit");
    fgets(buffer,BUFSIZ,stdin);
    num_bytes = write(fd,buffer,strlen(buffer));
    if(num_bytes == -1)
    {
        perror("Write");
        exit(EXIT_FAILURE);
    }
    if(strncmp(buffer,"exit",4) == 0)
    {
        break;
    }
}

close(fd);
unlink(FIFO_FILE);
return 0;
```

```
^G Help
^X Exit
```

```
^O Write Out
^R Read File
```

^W Where Is
^_ Replace

^K Cut
^U Paste

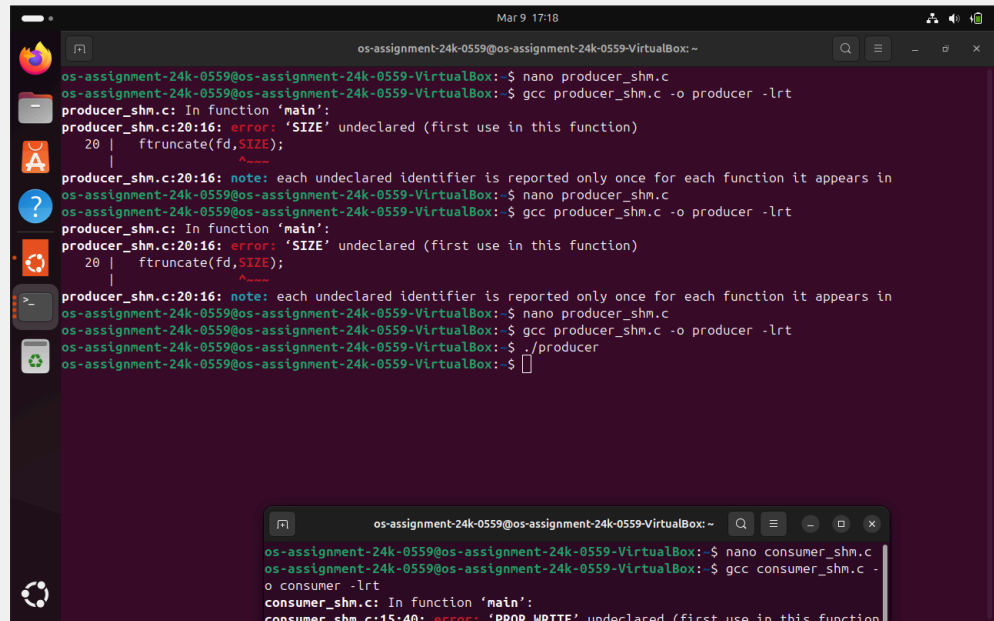
^T Execute
^J Justify

^C Lo
^/ Go

EXPERIMENT 6C).

IN LAB TASKS).

Ubuntu_Assignment [Running] - Oracle VirtualBox
File Machine View Input Devices Help



```
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano producer_shm.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc producer_shm.c -o producer -lrt
producer_shm.c: In function 'main':
producer_shm.c:20:16: error: 'SIZE' undeclared (first use in this function)
    20 |     ftruncate(fd,SIZE);
       |                ^~~~~
producer_shm.c:20:16: note: each undeclared identifier is reported only once for each function it appears in
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano producer_shm.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc producer_shm.c -o producer -lrt
producer_shm.c: In function 'main':
producer_shm.c:20:16: error: 'SIZE' undeclared (first use in this function)
    20 |     ftruncate(fd,SIZE);
       |                ^~~~~
producer_shm.c:20:16: note: each undeclared identifier is reported only once for each function it appears in
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano producer_shm.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc producer_shm.c -o producer -lrt
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./producer
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$

os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano consumer_shm.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc consumer_shm.c -o consumer -lrt
consumer_shm.c: In function 'main':
consumer_shm.c:15:40: error: 'PROD_WRITE' undeclared (first use in this function)
```

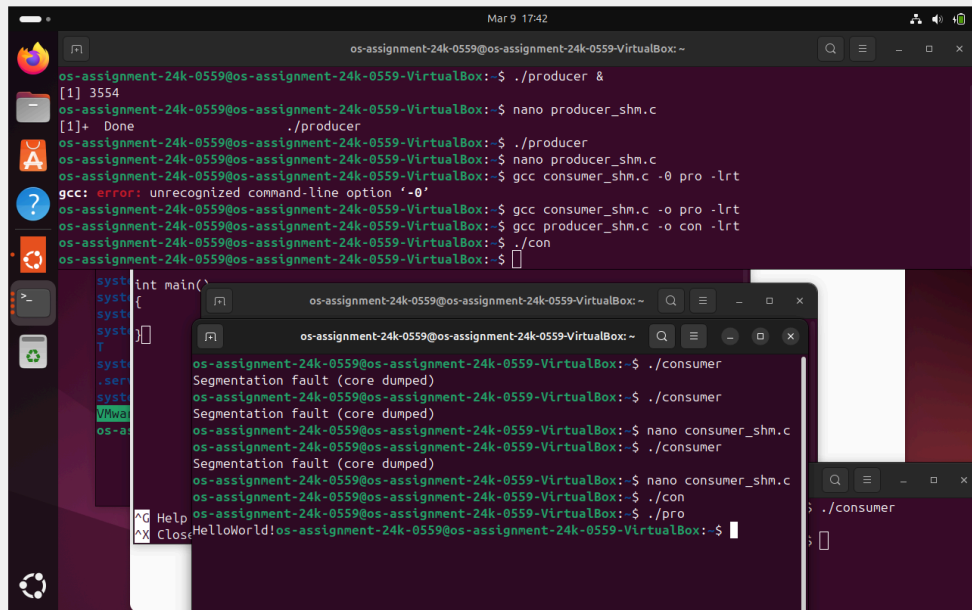
```
Mar 9 17:18
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: ~
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano producer_shm.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc producer_shm.c -o producer -lrt
producer_shm.c: In function 'main':
producer_shm.c:20:16: error: 'SIZE' undeclared (first use in this function)
   20 |     ftruncate(fd,SIZE);
      |                ^~~~~
producer_shm.c:20:16: note: each undeclared identifier is reported only once for each function it appears in
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano producer_shm.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc producer_shm.c -o producer -lrt
producer_shm.c: In function 'main':
producer_shm.c:20:16: error: 'SIZE' undeclared (first use in this function)
   20 |     ftruncate(fd,SIZE);
      |                ^~~~~
producer_shm.c:20:16: note: each undeclared identifier is reported only once for each function it appears in
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano consumer_shm.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc consumer_shm.c -o consumer -lrt
consumer_shm.c: In function 'main':
consumer_shm.c:15:40: error: 'PROT_WRITE' undeclared (first use in this function)
); did you mean 'PROT_READ'?
   15 |     ptr = (char *)mmap(0,SIZE,PROT_READ |PROT_WRITE,MAP_SHARED,fd,0);
      |                                     ^~~~~~
      |                                     PROT_WRITE
consumer_shm.c:15:40: note: each undeclared identifier is reported only once for each function it appears in
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano consumer_shm.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc consumer_shm.c -o consumer -lrt
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./consumer
Segmentation fault (core dumped)
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$
```

REASON:

THE REASON for it is that the consumer tries to open the shared memory before it exists.

So if the producer has not created the shared memory yet then the line fails the fd = shm_open(name,O_RDONLY, 0666);

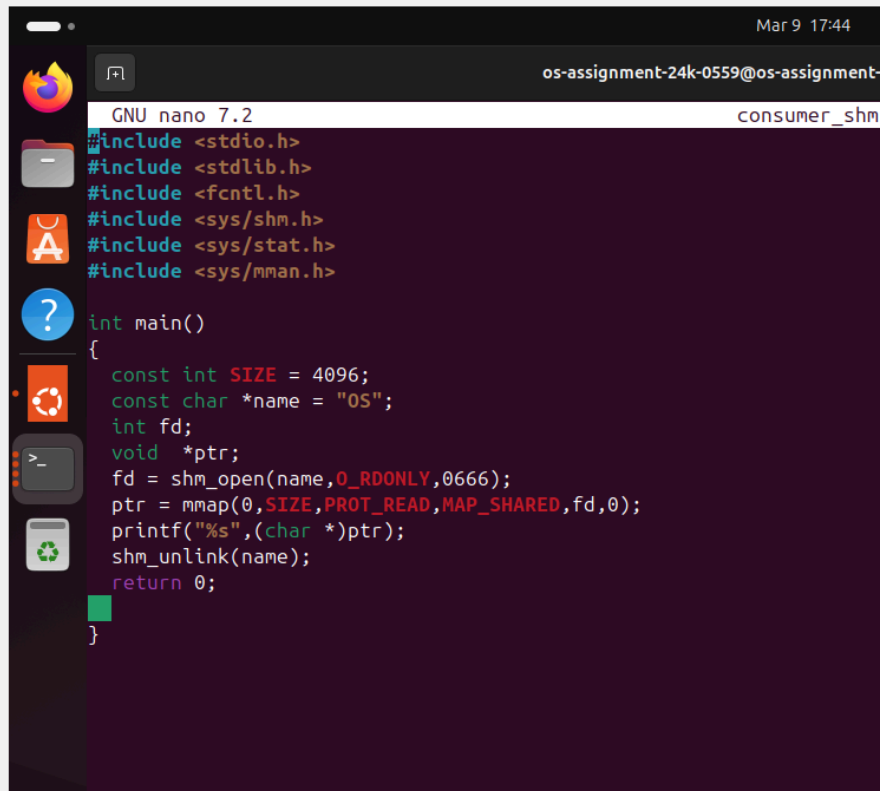
SO THAT'S WHY CONSUMER GETS AN ERROR.



```
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: ~
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./producer &
[1] 3554
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano producer_shm.c
[1]+  Done                  ./producer
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./producer
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano producer_shm.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc consumer_shm.c -o pro -lrt
gcc: error: unrecognized command-line option '-o'
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc consumer_shm.c -o pro -lrt
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ gcc producer_shm.c -o con -lrt
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./con
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$

int main()
{
    syst
    syst
    syst
    syst
    T
    syst
    .ser
    syst
    VMwa
    os-a
}

os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./consumer
Segmentation fault (core dumped)
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./consumer
Segmentation fault (core dumped)
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano consumer_shm.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./consumer
Segmentation fault (core dumped)
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ nano consumer_shm.c
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./con
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$ ./pro
HelloWorld!os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox:~$
```



The screenshot shows a terminal window titled "os-assignment-24k-0559@os-assignment-24k-0559" with a timestamp of "Mar 9 17:44". The terminal is running the GNU nano 7.2 text editor. The code being edited is a C program named "consumer_shm.c". The code includes headers for stdio, stdlib, fcntl, sys/shm, sys/stat, and sys/mman. The main function defines a constant SIZE of 4096, a character pointer name set to "0S", and an integer fd. It then calls shm_open with O_RDONLY, 0666 permissions, and fd. The pointer ptr is mapped using mmap with PROT_READ and MAP_SHARED flags. The program prints the contents of ptr using printf and then unlinks the shared memory object with shm_unlink before returning 0.

```
GNU nano 7.2 consumer_shm.c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <sys/shm.h>
#include <sys/stat.h>
#include <sys/mman.h>

int main()
{
    const int SIZE = 4096;
    const char *name = "0S";
    int fd;
    void *ptr;
    fd = shm_open(name, O_RDONLY, 0666);
    ptr = mmap(0, SIZE, PROT_READ, MAP_SHARED, fd, 0);
    printf("%s", (char *)ptr);
    shm_unlink(name);
    return 0;
}
```

```
Mar 9 17:45
os-assignment-24k-0559@os-assignment-24k-0559-VirtualBox: ~
GNU nano 7.2 producer_shm.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/shm.h>
#include <sys/stat.h>
#include <sys/mman.h>
#include <unistd.h>

int main()
{
    const int SIZE = 4096;
    const char *name = "05";
    const char *message_0 = "Hello";
    const char *message_1 = "World!";
    int fd;
    char *ptr;

    fd = shm_open(name, O_CREAT | O_RDWR, 0666);
    ftruncate(fd, SIZE);
    ptr = mmap(0, SIZE, PROT_WRITE, MAP_SHARED, fd, 0);
    sprintf(ptr, "%s", message_0);
    ptr += strlen(message_0);
    sprintf(ptr, "%s", message_1);

    return 0;
}
```